

Computational Social Science

Final Exam — Research Paper Guidelines

Academic Year: A.A. 2025–2026

Programme: MSc in Data Science — University of Trento

Lecturer: Gian Maria Campedelli, PhD

Session: Summer/Winter examination session

Overview

The final examination consists of a **research paper** submitted online during one of the dates of the summer or winter session. The paper must constitute an original piece of empirical social science research making use of computational methods. Students are evaluated on six dimensions, for a maximum total of **23 points**.

Students who received a midterm bonus will have it added to the final paper score. The resulting sum, rounded to the nearest integer, constitutes the final grade (capped at 30 *cum laude*). For example, a student who accumulated 5 bonus points from the written tests and receives 13 points out of 23 on the research paper obtains a final grade of 18.

A student may **decline the grade** of the final exam and attempt it again on one of the other available dates within the same session (retaining the points obtained in the written test(s)). If a student does not pass the exam in any of the summer/winter dates, they will forfeit all points accumulated through the written tests.

Evaluation Criteria and Point Allocation

Originality and Creativity carries the highest individual weight, reflecting its status as the most distinctive marker of genuinely independent scholarly work. Technically competent but intellectually generic papers will not score well overall.

Criterion	What is assessed	Points
1. Originality & Creativity	Does the paper ask a genuinely novel question or approach an existing problem from a fresh angle? Is the combination of methods and substantive domain innovative? Does the student go beyond replicating standard analyses?	6
2. Literature Review	Breadth and depth of engagement with relevant prior work. Appropriate source selection (rigorous, methodologically relevant). Evidence that the student commands the literature rather than merely listing it.	4
3. Research Design	Clarity and rigour of the research problem. Are the research questions and hypotheses stated precisely, testable, and aligned with the chosen methods? Is the design internally consistent?	4
4. Data & Methods	Appropriateness of data sources and methods for the research questions. Correct implementation and transparent reporting. Reproducibility, verified via code submission.	4
5. Interpretation of Results	Are findings correctly interpreted in light of the research questions? Does the student distinguish statistical from substantive significance, and association from causation?	3
6. Limitations	Honest, specific, and methodologically informed discussion of what the study cannot establish. Awareness of threats to validity, data constraints, and boundary conditions.	2
Total		23

Further notes on individual criteria

Originality & Creativity (6 pts). A high score requires more than applying a standard method to a new dataset. The paper should pose a genuinely interesting question, identify a meaningful gap, or combine methods and domains in a non-obvious way. Replication studies without a substantial methodological or theoretical extension will score low.

Literature Review (4 pts). Sources must be rigorous and methodologically relevant. Citing papers whose abstracts superficially match the topic, without engaging with their arguments, will not be considered sufficient. The review must *synthesise*, not merely list.

Research Design (4 pts). Research questions must be answerable with the proposed data and methods. Hypotheses must be falsifiable and directly linked to the design. Vague or internally inconsistent problem formulations will score low.

Data & Methods (4 pts). The description must be detailed enough to enable replication.

Scores on this criterion are partly verified empirically: 5 randomly selected papers per exam date will have their code executed to check that the results replicate.

Interpretation of Results (3 pts). Students must correctly distinguish statistical significance from substantive importance, and correlation from causation. Using causal language in observational designs that lack an appropriate identification strategy, for instance, will be penalised.

Limitations (2 pts). Generic disclaimers without methodological grounding are insufficient. This section must identify specific threats to validity and explain precisely why they constrain the conclusions. Students are also invited to reflect on avenues for future work and on how the analysis could be improved.

Format and Submission Requirements

Length

Minimum: 6,000 words. **Maximum:** 8,000 words.

The word count **excludes** the reference list. All other sections — including the abstract, table captions, and figure captions — are counted.

These limits will not be enforced rigidly, but papers that are extremely short (e.g. 4 pages) or excessively long (e.g., 50 pages) will be penalized.

Template and typesetting

Students may use **any editor** — including L^AT_EX/Overleaf, Microsoft Word, or Google Docs — provided the output is professional and consistently formatted. I would advise using L^AT_EX, but you are free to lose your mind over a Word document if you prefer.

A legible font (not Calibri, please) of at least 11pt, standard margins (2–2.5 cm), and line spacing of 1.15 or greater are required.

Figures and plots should be submitted in **vectorised formats (PDF, SVG, or EPS)** wherever possible to ensure high resolution. PNG is acceptable when vector output is unavailable.

Tables and visualisations are **fundamental**. They must be numbered, captioned, and referenced in the text.

The paper must be written in **English**. Colloquial phrasing, undefined jargon, and stylistic sloppiness will be penalised.

Papers submitted without formal exam registration **will be rejected outright**.

Required structure

Minor variations in section labelling are acceptable (e.g. splitting Results into subsections, or merging Discussion and Conclusions). Substantial structural deviations require prior approval.

Section	Content
Title, author, abstract	Title of the paper, full name and student ID, and a self-contained abstract of 150–250 words summarising the research question, method, main findings, and implications.
Introduction	Contextualises the research problem, motivates its relevance, and previews the structure of the paper.
Background / Related Work	Critical engagement with prior literature. Not a list of summaries, but a synthesis that identifies gaps and positions the paper’s contribution.
Research Problem	Explicit formulation of the research question(s) and, where appropriate, testable hypotheses. These must follow directly from the literature review.
Analytical Approach	Description of data sources (provenance, coverage, preprocessing) and methods (design, estimation strategy, software). Sufficient detail to enable replication.
Results	Presentation of findings with tables and figures where appropriate. Interpretation anchored strictly to what the data and method support.
Limitations	Specific, methodologically grounded discussion of what the study cannot establish and why.
Discussion & Conclusions	Synthesis of findings, implications for theory or policy, and directions for future research.
References	Complete and consistently formatted bibliography (e.g., APA, Harvard, Nature).

References

Any consistent reference style is accepted: **APA, Harvard, or Nature**.

All cited sources must exist and be retrievable. The use of a reference manager such as **Zotero** is strongly recommended.

Hallucinated references — papers that do not exist, have incorrect titles, wrong authors, or fictitious publication details — will result in the **immediate rejection of the paper**, regardless of the quality of the remaining content. This is a hard rule with no exceptions as it signals the substantial use of LLMs possibly beyond the permitted use (see below).

Code and data submission

Together with the paper, students must submit a **zipped folder** containing: (a) all code used for data processing and analysis; (b) the data, or clear instructions for accessing it if it is too large or subject to access restrictions; and (c) a short **README** file explaining how to reproduce the main results.

Reproducibility check: in each session, 5 papers will be selected at random and their

code will be executed. If the code does not run without errors, or does not reproduce the results reported in the paper, the score on the Data & Methods criterion will be penalised accordingly.

Code must be clearly commented.

Research Topic and Methodological Scope

Type of Research

Students are free to choose the type of research their paper pursues: it can be descriptive, causal, correlational, or predictive. What matters is the creativity and analytical depth of the work.

Descriptive papers are accepted, but descriptive does not mean computing standard summary statistics. It means providing a specialised exploratory analysis that illuminates primary data collected by the student, or large datasets that are difficult to access and process. A superficial descriptive analysis of easily available secondary data will not be accepted.

Students are encouraged to integrate multiple perspectives — for example, combining a descriptive and a causal component.

Topic

Students are free to choose any research problem from the social sciences, provided it constitutes a genuine application of computational approaches to a social science question. Substantive domains include, but are not limited to: sociology, political science, economics, demography, criminology, public health, education, and communication studies.

Applications to purely biological, chemical, or physical problems will not be accepted, even if computational methods are involved. The substantive question must be social scientific in nature.

Methods

Students may use **any appropriate method**, including methods not covered during the course. Examples of approaches that are permitted but were not treated in the course include:

Agent-based modelling (ABM)

Self-exciting processes and point processes (e.g. Hawkes processes)

Complex systems models based on ordinary or partial differential equations (ODEs/PDEs)

Structural models from economics or sociology

Deep learning and computer vision approaches

Reinforcement learning

Students are encouraged to **integrate multiple approaches** where coherent and well-motivated — for example, combining a causal identification strategy with a predictive task, or an in-depth descriptive analysis with an inferential one. Such integrations will score highly on the Originality criterion.

Policy on the Use of Large Language Models

The use of large language models (LLMs) is **permitted** for spell-checking, proofreading, stylistic polishing, and assistance with data analysis. Students are not required to disclose LLM use.

The paper will be treated as substantially generated by an LLM — and **rejected outright** — if either of the following is detected:

Prompt injection artefacts. Fragments that are clear by-products of unedited LLM output: visible prompt text, meta-commentary such as “here is a possible introduction”, placeholders such as “[insert citation here]”, or structurally generic language that does not engage with the specific content of the paper.

Hallucinated references. Any cited source that does not exist, has an incorrect title, lists wrong authors, or contains fabricated publication details. This is the most reliable signal of LLM over-reliance and is treated as academic dishonesty. Papers containing hallucinated references are **immediately rejected**, regardless of the quality of the rest of the work.

Given that LLMs are widely available and that detection software is unreliable, the evaluation places correspondingly greater weight on **originality, methodological rigour, and intellectual depth** — qualities that are difficult to produce with generic LLM output alone. A technically correct but intellectually thin paper will score poorly on Criteria 1–3 even if it performs well on Criteria 4–6.

Policy on Plagiarism

Plagiarism will be checked using the university’s internal software. Papers found to contain substantial portions of plagiarised text will be rejected.

Heartfelt Suggestions

Writing a research paper requires considerable craft. For most of you, this will be your first experience of it, and you will likely find it challenging at first — that is entirely normal. The single most effective way to prepare, beyond studying the course materials, is to read, read, *read*. Find articles relevant to your domain of interest and study how professional scientists develop a narrative, visualise patterns, and communicate results. The more you read, the more naturally you will find ideas, assess their originality, and translate them into writing.

Think of this paper as a first step toward your Master’s thesis. If you are drawn to research, treat it as your first practical encounter with the scientific process — with all the difficulty, and all the satisfaction, that entails. Try to have fun. Without fun all is lost.

Submission Checklist

Before submitting, verify that each of the following conditions is met.

- ☐ The paper is in English and within the 6,000–8,000 word limit (references excluded).
 - ☐ All required sections are present and clearly labelled.
 - ☐ The reference list is complete, accurate, and formatted in a consistent style.
 - ☐ All cited sources have been verified to exist and are correctly attributed.
 - ☐ Code and data are zipped and submitted together with the paper.
 - ☐ A README file explains how to reproduce the main results.
 - ☐ Figures are in vectorised format (PDF/SVG) where possible, numbered, and captioned.
 - ☐ The document is professionally typeset, free of colloquial language, and consistently formatted.
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Questions regarding these guidelines should be directed to the course lecturer before the submission deadline.